PRECISION IT SYSTEMS • OPERATIONS MANUAL

Admin-to-Technician Project Workflow, Verification & Completion Guide

Version: 2.4 (Production)
Updated: 27 Oct 2026
Modules: Projects, Calendar, Tech Portal, Wages

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Create Project	**Approve & Rate**	**Delegate Tasks**	**Accept / Reject**	**Execute & Proof**	**Verify & Pay**
Admin / PM	Admin	Admin / PM	Technician	Technician	Admin

STAGE 1: PROJECT INITIALIZATION & BOM BUILDER (ADMIN)

Attribute	Technical Details & Routing	Example Live Payload / Values
Action & Route	`POST /projects` (or wizard: `POST /projects/wizard`) Method: `ProjectController::store()`	Creates new master installation project record
Core Fields	• Auto-sequential Project Code generator • Type: CCTV, Attendance, Networking, Hybrid • Assigns Lead Technician (FK: `users.id`)	`project_code`: **PRJ-2026-0002** `title`: **A.C.S. Medical College - CCTV & Access** `lead_technician_id`: **3 (Bob Miller)** `budget`: **?1,50,000.00**
BOM (Bill of Materials)	Inline array in `ProjectMaterial` table. Auto-learns products and vendors by normalized name key.	• 16x Hikvision 4MP Dome Camera (Warehouse) • 3x Cat6 UTP 305m Cable Bundle (Warehouse) • 1x 16-Channel 4K NVR + 4TB Surveillance HDD
Initial State	Sets `status = 'pending_approval'` (or `approved` if created directly by Admin with company).	`PENDING_APPROVAL` ? Logs in `ProjectAuditLog`

STAGE 2: ADMIN APPROVAL & PER-METRE CABLING RATE SETUP

Item	Backend Logic	Financial / Operational Impact
Action & Route	`POST /projects/{project}/approve` Method: `ProjectController::approve()`	Restricted strictly to administrators (`abort_unless(isAdmin(), 403)`)
Parameters	• `company_id`: Executing client entity • `per_metre_rate`: Baseline cabling payout rate • `approved_value`: Agreed budget ceiling • `po_number`: Client Purchase Order	`company_id`: **2 (ACS Medical College)** `per_metre_rate`: **?12.00 / metre** `approved_value`: **?1,45,000.00** `po_number`: **PO-2026-MED-89**
State Transition	`status = 'approved'`, `approved_by = Admin ID`, `approved_on = now()`.	`APPROVED` (Project now eligible for active task dispatching)

STAGE 3: TASK DELEGATION & CHECKLIST GENERATION (ADMIN / PM)

Step	Method & Routing	Output Data & Auto-Notification
A. Workflow Template	`POST /projects/{project}/tasks/apply-template` `ProjectTaskController::applyTemplate()`	Auto-generates standard CCTV installation phases (Conduit, Cabling, Mount, IP Config, Testing).
B. Custom Task Creation	`POST /calendar/tasks` or `POST /projects/{project}/tasks` `CalendarController::storeTask()`	`title`: **Pull 110m Cat6 Cable & Mount Cams 1-4** `assigned_to`: **3 (Bob Miller)** | `priority`: **urgent** `shift`: **Day Shift** | `due_date`: **2026-10-27**
C. Checklist Creation	Inline array or `POST /project-tasks/{task}/checklists`	1. Lay conduit along 2nd-floor ceiling 2. Pull 110m Cat6 bundle from rack to corridor 3. Terminate RJ45 keystones & test wiremap 4. Securely mount 4 dome cameras
D. Alert Notification	`NotificationLog::create(['channel' => 'whatsapp', ...])`	Dispatches dashboard bell notification and generates formatted WhatsApp task briefing link.

STAGE 4: TECHNICIAN INSPECTION, ACCEPTANCE & REJECTION FLOW

? Privacy & Data Sanitization (Server-Side Enforced):
When a Technician or Worker opens their tasks via `CalendarController::index(['tab' => 'my_tasks'])`, the server passes task models through `sanitizeTaskForWorker()`. Sensitive client financial margins, master project budgets, vendor procurement rates, and BOM cost structures are **completely stripped out**. Technicians see only technical scope, date, shift, location, and checklists.

Technician Action	Code Execution & Route	Database Updates & Audit Log
Path A: Accept Task (Normal Flow)	`POST /calendar/tasks/{task}/accept` Method: `CalendarController::acceptTask()` Security check: `$task->assigned_to === Auth::id()`	• `status = 'in_progress'` • `accepted_at = now()` • Clears any previous rejection notes • Audit: *"Bob Miller accepted task 'Pull 110m Cat6...'"* IN_PROGRESS
Path B: Reject Task (Site Inaccessible)	`POST /calendar/tasks/{task}/reject` Method: `CalendarController::rejectTask()` Requires reason (min 3 chars).	• `status = 'rejected'` • `rejection_reason = "Site locked; painting in progress"` • `rejected_at = now()`, `rejected_by = Auth::id()` • Routes task back to Admin with warning badge for reassignment. REJECTED

STAGE 5: FIELD EXECUTION, CHECKLIST TICKING & PHOTOGRAPHIC VERIFICATION

Field Activity	Backend Route & Controller	Verification Artifacts Recorded
1. Interactive Checklist	`POST /project-tasks/checklists/{checklist}/toggle` `ProjectTaskController::toggleChecklist()`	• Toggles `is_completed = true/false` • Recalculates `task.checklist_progress` (e.g. 50% ? 75% ? 100%)
2. Punch-in Selfie & Output Proof	`POST /projects/{project}/work-days/{workDay}/attendances` `WorkDayController::storeAttendance()`	• `photo_path`: Live webcam/phone punch selfie • `cabling_metres`: **110.0 m** • `cameras_installed`: **4 cameras** • `attendance_type`: **Full Day (?900.00 base)**
3. Material Consumption	`POST /projects/{project}/materials/consume` `ProjectMaterialController::consume()`	Deducts physical stock from project BOM and creates `TaskMaterialConsumption` ledger entry.
4. Site Blockers (If Any)	`POST /projects/{project}/tasks/{task}/flag-blocker` `ProjectTaskController::flagBlocker()`	Marks `status = 'blocked'`, logs reason, and fires automated WhatsApp alert to project managers.

STAGE 6: ADMIN REVIEW, DONE VERIFICATION & WAGE SETTLEMENT

? 3-Point Admin Verification Checklist Before Sign-Off:
1. **Task Board Audit:** All project tasks must reach `completed` status (`taskCompletionPercentage == 100%`).
2. **Visual Proof Inspection:** Admin opens the **Shifts & Photos Lightbox** to inspect the technician's check-in punch selfie.
3. **Material & Output Reconciliation:** Verified cabling metres pulled (110m) matches warehouse Cat6 cable depletion.

Final Sign-off Step	Action & Route	Outcome & Data Results
A. Mark Task Done	`PATCH /projects/{project}/tasks/{task}/status` `ProjectTaskController::updateStatus()`	Sets `status = 'completed'`, `completed_at = now()`, `completed_by = Auth::id()`.
B. Project Closeout	`PATCH /projects/{project}/status` `ProjectController::updateStatus()`	Sets `status = 'completed'`, `progress_percentage = 100`, `completed_at = today()`.
C. Automated Wage Payout	`POST /wages/settle` `WageController::settle()`	Formula: `(Daily Shift Base: ?900) + (110m Cabling @ ?12: ?1,320) - (Advances) = Net ?2,220.00` Disbursed via Cash or UPI and marked ? SETTLED in Weekly Wages.

DEVELOPER & OPERATIONAL QUICK-REFERENCE CHEAT SHEET

Operation	HTTP Route	Controller & Method	Authorized Role	Resulting Status
Create Project	`POST /projects`	`ProjectController@store`	Admin, Staff	PENDING_APPROVAL
Approve Project	`POST /projects/{id}/approve`	`ProjectController@approve`	Admin Only	APPROVED
Create Project Task	`POST /calendar/tasks`	`CalendarController@storeTask`	Admin, PM	PENDING
Accept Task	`POST /calendar/tasks/{id}/accept`	`CalendarController@acceptTask`	Assigned Tech	IN_PROGRESS
Reject Task	`POST /calendar/tasks/{id}/reject`	`CalendarController@rejectTask`	Assigned Tech	REJECTED
Toggle Checklist	`POST /project-tasks/checklists/{id}/toggle`	`ProjectTaskController@toggleChecklist`	Assigned Tech	`is_completed = true`
Complete Task	`PATCH /projects/{id}/tasks/{task}/status`	`ProjectTaskController@updateStatus`	Tech / Admin	COMPLETED
Close Project	`PATCH /projects/{id}/status`	`ProjectController@updateStatus`	Admin Only	COMPLETED (100%)

Operation	HTTP Route	Controller & Method	Authorized Role	Resulting Status
Settle Wages	`POST /wages/settle`	`WageController@settle`	Admin Only	**? SETTLED**